

Will two blues always fit?

Name: _____

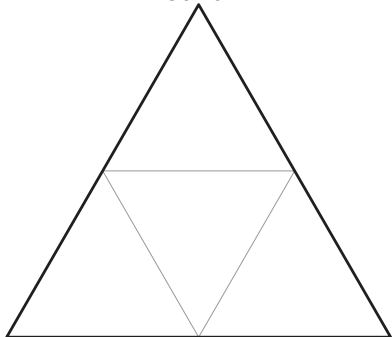
Date: _____

Use only small ($1\times$) blue rhombi. Each blue must cover two whole triangle spaces. No overlaps or pieces outside.

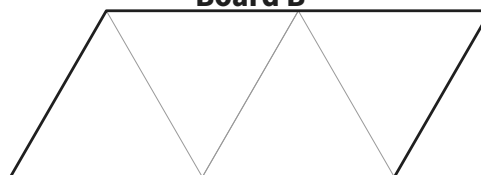
1. Same area. Same answer?

Each outline has four triangle spaces. Try to fill each with two blues.

Board A



Board B



Which board can you fill? Show a filling or explain a difficulty.

—|—
One small block edge

Adult: print **100% / Actual Size**, single-sided. Default edge: 1 inch (25.4 mm). Compare with a $1\times$ green triangle before using these mats.

Leave the fewest gaps

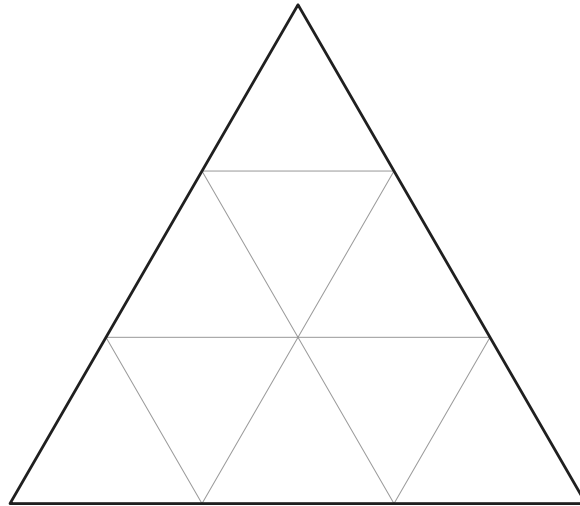
Name: _____

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Use small blue rhombi on pairs of whole triangle spaces. Leave as few spaces empty as you can. No overlaps or pieces outside.

2. A bigger triangle.

Try a packing. Change it if you can leave fewer gaps.



My best packing leaves _____ triangle spaces empty.

Show why nobody can leave fewer.

Go further: Predict the fewest gaps for a triangle with four small edges on each side. Explain your prediction.

Build first. Explain by pointing, talking, or drawing.

Can new pieces help?

Name: _____

Date: _____

3. Let purple join the blues.

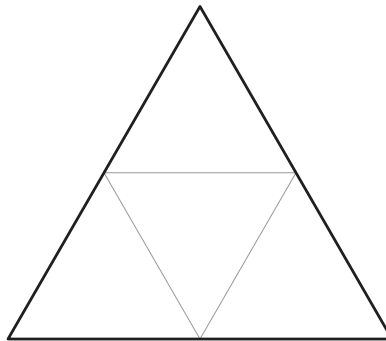
Return to the big triangle on page 2. Add purple chevrons to your small blues. Can you leave fewer triangle spaces empty?

Can purple help? Why or why not?

4. Now try two pink triangles.

Set the blues and purples aside. Try two pink right triangles in Board A below. You may turn them over.

Board A again



What changed? Why does our earlier reason no longer apply?

Go further: On the big triangle from page 2, compare two goals: use the fewest green triangles to fill the gaps, or use the fewest blocks altogether. Does purple help either goal?

Build first. Explain by pointing, talking, or drawing.